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Exploratory Structural Equation Modeling (ESEM)

Readings: Alamer & Marsh (2022); Marsh et al. (2014)

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```
library(lavaan)
library(semTools)
library(modelsummary)
library(tinytable)
library(dplyr)
knitr::read_chunk("R/esem.R")
```

While confirmatory factor analysis (CFA) has been the standard approach for validating measurement models, for some constructs, items may not perfectly measure only one intended factor. For example, personality items usually capture more than one dimensions, like some agreeableness items may also reflect conscientiousness. However, CFA generally assumes a **simple structure** where each item loads on only one factor, or at least that there are only a few cross-loadings that can be specified a priori. When many items have small but non-negligible cross-loadings, CFA can produce poor model fit and inflated factor correlations, leading to distorted conclusions about the underlying constructs.

CFA Loading Matrix:

$$\begin{bmatrix} \lambda_{11} & 0 \\ \lambda_{21} & 0 \\ \lambda_{31} & 0 \\ \lambda_{41} & 0 \\ \lambda_{51} & 0 \\ \lambda_{61} & 0 \\ 0 & \lambda_{72} \\ 0 & \lambda_{82} \\ 0 & \lambda_{92} \\ 0 & \lambda_{10,2} \\ 0 & \lambda_{11,2} \\ 0 & \lambda_{12,2} \end{bmatrix}$$

Instead, **Exploratory Factor Analysis (EFA)**, or some authors calling it **Exploratory Structural Equation Modeling (ESEM)** when specified within a SEM framework, allows all items to load on all factors, with the expectation that most cross-loadings will be small. Because of the allowance for cross-loadings, EFA can sometimes provide a more accurate representation of the data, better model fit, and more realistic factor correlations. Doing EFA within an SEM framework allows us to retain the advantages of SEM, such as fit indices and the ability to use full information maximum likelihood (FIML) estimation for handling missing data.

EFA Loading Matrix (Bold values indicate primary loadings):

$$\begin{bmatrix} \boldsymbol{\lambda}_{11} & \approx 0 \\ \lambda_{21} & \approx 0 \\ \lambda_{31} & \approx 0 \\ \lambda_{41} & \approx 0 \\ \lambda_{51} & \approx 0 \\ \lambda_{61} & \approx 0 \\ \approx 0 & \lambda_{72} \\ \approx 0 & \lambda_{82} \\ \approx 0 & \lambda_{92} \\ \approx 0 & \lambda_{10,2} \\ \approx 0 & \lambda_{11,2} \\ \approx 0 & \lambda_{12,2} \end{bmatrix}$$

Dataset: L2 Passion Scale

Alamer & Marsh (2022) applied ESEM to the **L2 Passion Scale**, an adaptation of the Passion Scale (Vallerand et al., 2003) for the language-learning context. The scale measures two types of passion based on the **Dualistic Model of Passion (DMP)**:

Factor	Items	Description
HP: Harmonious Passion	HP1–HP6	Autonomous internalization; passion integrated with other life activities
OP: Obsessive Passion	OP1–OP6	Controlled internalization; uncontrollable urge to engage in the activity

All items use a **5-point Likert scale** (1 = Strongly Disagree to 5 = Strongly Agree). The dataset contains N = 214 Saudi EFL undergraduate students.

> ℹ Data Access
>
> Data are available from the IRIS Database: https://www.iris-database.org/details/WYMmh-Qz9eK. The Excel file (`Passion dataset S1- Alamer & Marsh 2022.xlsx`) is saved in the `data/` folder of this project.

Descriptive Statistics

```
modelsummary::datasummary_skim(dplyr::select(study1, -Gender))
```

	Unique	Missing Pct.	Mean	SD	Min	Median	Max	Histogram
HP1	4	0	4.5	0.6	2.0	5.0	5.0	
HP2	4	0	4.5	0.7	2.0	5.0	5.0	
HP3	6	0	4.0	0.9	1.0	4.0	5.0	
HP4	4	0	4.6	0.6	2.0	5.0	5.0	
HP5	4	0	4.3	0.8	2.0	4.0	5.0	
HP6	5	0	3.9	0.9	1.0	4.0	5.0	
OP1	6	0	3.1	1.1	1.0	3.0	5.0	
OP2	6	0	3.3	1.1	1.0	3.0	5.0	
OP3	6	1	3.6	1.2	1.0	4.0	5.0	
OP4	6	0	2.4	1.2	1.0	2.0	5.0	
OP5	6	1	2.6	1.1	1.0	2.0	5.0	
OP6	6	0	2.7	1.1	1.0	3.0	5.0	

Inter-item Correlations

```
items <- c("HP1", "HP2", "HP3", "HP4", "HP5", "HP6",
           "OP1", "OP2", "OP3", "OP4", "OP5", "OP6")
cor_s1 <- cor(study1[items], use = "pairwise.complete.obs")
bg <- rgb(as.integer(cor_s1 > 0), 0, as.integer(cor_s1 < 0),
          alpha = abs(cor_s1))
bg[upper.tri(cor_s1, diag = TRUE)] <- NA
nv <- length(items)
datasummary_correlation(study1[items]) |>
  style_tt(i = 1:nv, j = 1:nv + 1, background = bg)
```

	HP1	HP2	HP3	HP4	HP5	HP6	OP1	OP2	OP3	OP4	OP5	OP6
HP1	1	.	.	.	.	.	.	.	.	.	.	.
HP2	.27	1	.	.	.	.	.	.	.	.	.	.
HP3	.25	.51	1	.	.	.	.	.	.	.	.	.
HP4	.34	.54	.46	1	.	.	.	.	.	.	.	.
HP5	.30	.34	.36	.44	1	.	.	.	.	.	.	.
HP6	.31	.22	.40	.26	.42	1	.	.	.	.	.	.
OP1	-.03	-.08	.01	-.14	-.22	.05	1	.	.	.	.	.
OP2	.02	.05	.11	.00	-.13	.04	.72	1	.	.	.	.
OP3	.20	.29	.28	.28	.20	.21	.13	.28	1	.	.	.
OP4	.02	-.00	.12	-.01	.02	.03	.23	.25	.22	1	.	.
OP5	.05	.22	.30	.19	.12	.17	.13	.18	.38	.37	1	.
OP6	.04	.21	.14	.15	.04	.10	.22	.27	.44	.36	.63	1

Note

Blue cells = positive correlations, red = negative, intensity = magnitude. Items within the same subscale should cluster together; HP items (HP1–HP6) and OP items (OP1–OP6) should form two clearly separated blocks.

EFA vs. CFA

The Problem with CFA for Multidimensional Scales

In standard CFA, every cross-loading is **fixed to zero**. When items genuinely share some degree of similarity across conceptually related constructs (as is common in psychology), fixing these small cross-loadings to zero forces their variance into the **factor correlation**, artificially inflating it.

For example, Alamer & Marsh (2022) found that the HP–OP correlation was $r = .34$ in CFA but only $r = .23$ in EFA—a reduction of .11, entirely attributable to variance expressed through cross-loadings in EFA but absorbed into the factor correlation in CFA.

What ESEM Does Differently

To be consistent with the paper, subsequently we use ESEM for the type of EFA within the SEM framework and can have specific loadings targetted to zero. We refer to EFA the traditional technique with non-iterative methods of estimation and rotation. ESEM integrates the advantages of EFA and CFA:

Feature	EFA	CFA	ESEM
Cross-loadings	All estimated	All fixed to zero	Freely estimated, targeted toward zero by rotation
Goodness-of-fit indices	✗	✓	✓
Structural paths	✗	✓	✓
Measurement invariance	✗	✓	✓
Factor correlations	Determined by rotation	Directly estimated	Directly estimated

In ESEM, **target rotation** (or Geomin) is used to achieve near-simple structure: cross-loadings are targeted toward zero but not forced there, so small genuine cross-loadings can be freely estimated.

Specifying ESEM in `lavaan`

The `efa()` block inside a `cfa()` model string marks a set of factors as belonging to the same EFA block, causing all items in that block to load on all factors:

```
# ESEM: both factors estimated in one efa() block with geomin rotation.
# Cross-loadings are freely estimated (targeted toward zero by rotation).
esem_model <- "
  efa('esem')*HP + efa('esem')*OP =~
    HP1 + HP2 + HP3 + HP4 + HP5 + HP6 +
    OP1 + OP2 + OP3 + OP4 + OP5 + OP6
  OP1 ~~ OP2   # correlated uniqueness (OP1 and OP2)
"
target_mat <- matrix(NA, nrow = 12, ncol = 2)
target_mat[1:6, 2] <- 0  # HP items should not load on OP factor
target_mat[7:12, 1] <- 0 # OP items should not load on HP factor
fit_esem <- cfa(
    esem_model,
    data = study1,
    estimator = "MLR",
    rotation = "target",
```

```
    rotation.args = list(target = target_mat)
)
summary(fit_esem, fit.measures = TRUE, standardized = TRUE)
```

- All 12 items load on **both** HP and OP; target rotation pushes cross-loadings toward zero.
- The model retains CFA features: fit indices, factor correlations, structural extensions.

Example: L2 Passion Scale

Example 1: CFA

```
cfa_model <- "
  HP =~ HP1 + HP2 + HP3 + HP4 + HP5 + HP6
  OP =~ OP1 + OP2 + OP3 + OP4 + OP5 + OP6
  OP1 ~~ OP2   # correlated uniqueness (OP1 and OP2 share wording)
"
fit_cfa <- cfa(cfa_model, data = study1,
               std.lv    = TRUE,
               estimator = "MLR")
summary(fit_cfa, fit.measures = TRUE, standardized = TRUE)
```

```
lavaan 0.6-21 ended normally after 29 iterations

  Estimator                                         ML
  Optimization method                           NLMINB
  Number of model parameters                        26

                                                   Used       Total
  Number of observations                           210         214

Model Test User Model:
                                              Standard      Scaled
  Test Statistic                               114.941     110.724
  Degrees of freedom                                52          52
  P-value (Chi-square)                           0.000       0.000
  Scaling correction factor                                  1.038
    Yuan-Bentler correction (Mplus variant)

Model Test Baseline Model:

  Test statistic                               776.960     685.670
  Degrees of freedom                                66          66
  P-value                                        0.000       0.000
  Scaling correction factor                                  1.133
```

```
User Model versus Baseline Model:

  Comparative Fit Index (CFI)                    0.911       0.905
  Tucker-Lewis Index (TLI)                       0.888       0.880

  Robust Comparative Fit Index (CFI)                         0.913
  Robust Tucker-Lewis Index (TLI)                            0.890

Loglikelihood and Information Criteria:

  Loglikelihood user model (H0)              -3049.913   -3049.913
  Scaling correction factor                                  1.239
      for the MLR correction
  Loglikelihood unrestricted model (H1)      -2992.442   -2992.442
  Scaling correction factor                                  1.105
      for the MLR correction

  Akaike (AIC)                                6151.825    6151.825
  Bayesian (BIC)                              6238.850    6238.850
  Sample-size adjusted Bayesian (SABIC)       6156.467    6156.467

Root Mean Square Error of Approximation:

  RMSEA                                          0.076       0.073
  90 Percent confidence interval - lower         0.057       0.055
  90 Percent confidence interval - upper         0.095       0.092
  P-value H_0: RMSEA <= 0.050                    0.013       0.021
  P-value H_0: RMSEA >= 0.080                    0.377       0.291

  Robust RMSEA                                               0.075
  90 Percent confidence interval - lower                     0.055
  90 Percent confidence interval - upper                     0.094
  P-value H_0: Robust RMSEA <= 0.050                         0.020
  P-value H_0: Robust RMSEA >= 0.080                         0.342

Standardized Root Mean Square Residual:

  SRMR                                           0.080       0.080

Parameter Estimates:

  Standard errors                             Sandwich
  Information bread                           Observed
  Observed information based on                Hessian

Latent Variables:
                   Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
  HP =~
```

```
    HP1               0.285    0.052    5.529    0.000    0.285    0.439
    HP2               0.492    0.053    9.264    0.000    0.492    0.698
    HP3               0.649    0.054   12.053    0.000    0.649    0.696
    HP4               0.435    0.046    9.496    0.000    0.435    0.717
    HP5               0.447    0.060    7.504    0.000    0.447    0.583
    HP6               0.456    0.070    6.552    0.000    0.456    0.484
  OP =~
    OP1               0.270    0.099    2.724    0.006    0.270    0.236
    OP2               0.368    0.091    4.026    0.000    0.368    0.327
    OP3               0.655    0.078    8.439    0.000    0.655    0.551
    OP4               0.549    0.102    5.375    0.000    0.549    0.455
    OP5               0.817    0.076   10.703    0.000    0.817    0.771
    OP6               0.871    0.068   12.736    0.000    0.871    0.806

Covariances:
                   Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
 .OP1 ~~
   .OP2               0.821    0.097    8.491    0.000    0.821    0.694
  HP ~~
    OP                0.345    0.080    4.326    0.000    0.345    0.345

Variances:
                   Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
   .HP1               0.339    0.042    8.017    0.000    0.339    0.807
   .HP2               0.255    0.044    5.743    0.000    0.255    0.513
   .HP3               0.449    0.086    5.219    0.000    0.449    0.516
   .HP4               0.179    0.033    5.376    0.000    0.179    0.486
   .HP5               0.387    0.068    5.728    0.000    0.387    0.660
   .HP6               0.680    0.110    6.153    0.000    0.680    0.765
   .OP1               1.233    0.101   12.254    0.000    1.233    0.944
   .OP2               1.134    0.093   12.146    0.000    1.134    0.893
   .OP3               0.984    0.099    9.961    0.000    0.984    0.696
   .OP4               1.156    0.136    8.505    0.000    1.156    0.793
   .OP5               0.456    0.092    4.952    0.000    0.456    0.406
   .OP6               0.411    0.084    4.909    0.000    0.411    0.351
    HP                1.000                               1.000    1.000
    OP                1.000                               1.000    1.000
```

Example 2: ESEM

```
# ESEM: both factors estimated in one efa() block with geomin rotation.
# Cross-loadings are freely estimated (targeted toward zero by rotation).
esem_model <- "
  efa('esem')*HP + efa('esem')*OP =~
    HP1 + HP2 + HP3 + HP4 + HP5 + HP6 +
    OP1 + OP2 + OP3 + OP4 + OP5 + OP6
  OP1 ~~ OP2   # correlated uniqueness (OP1 and OP2)
"
```

```
target_mat <- matrix(NA, nrow = 12, ncol = 2)
target_mat[1:6, 2] <- 0  # HP items should not load on OP factor
target_mat[7:12, 1] <- 0 # OP items should not load on HP factor
fit_esem <- cfa(
    esem_model,
    data = study1,
    estimator = "MLR",
    rotation = "target",
    rotation.args = list(target = target_mat)
)
summary(fit_esem, fit.measures = TRUE, standardized = TRUE)
```

```
lavaan 0.6-21 ended normally after 35 iterations

  Estimator                                         ML
  Optimization method                           NLMINB
  Number of model parameters                        38
  Row rank of the constraints matrix                 2

  Rotation method                          PST OBLIQUE
  Rotation algorithm (rstarts)                GPA (30)
  Standardized metric                             TRUE
  Row weights                                     None

                                                   Used       Total
  Number of observations                           210         214

Model Test User Model:
                                              Standard      Scaled
  Test Statistic                                79.092      75.739
  Degrees of freedom                                42          42
  P-value (Chi-square)                           0.000       0.001
  Scaling correction factor                                  1.044
    Yuan-Bentler correction (Mplus variant)

Model Test Baseline Model:

  Test statistic                               776.960     685.670
  Degrees of freedom                                66          66
  P-value                                        0.000       0.000
  Scaling correction factor                                  1.133

User Model versus Baseline Model:

  Comparative Fit Index (CFI)                    0.948       0.946
  Tucker-Lewis Index (TLI)                       0.918       0.914
```

```
  Robust Comparative Fit Index (CFI)                         0.950
  Robust Tucker-Lewis Index (TLI)                            0.921

Loglikelihood and Information Criteria:

  Loglikelihood user model (H0)              -3031.988   -3031.988
  Scaling correction factor                                  1.176
      for the MLR correction
  Loglikelihood unrestricted model (H1)      -2992.442   -2992.442
  Scaling correction factor                                  1.105
      for the MLR correction

  Akaike (AIC)                                6135.977    6135.977
  Bayesian (BIC)                              6256.473    6256.473
  Sample-size adjusted Bayesian (SABIC)       6142.404    6142.404

Root Mean Square Error of Approximation:

  RMSEA                                          0.065       0.062
  90 Percent confidence interval - lower         0.042       0.039
  90 Percent confidence interval - upper         0.087       0.083
  P-value H_0: RMSEA <= 0.050                    0.128       0.178
  P-value H_0: RMSEA >= 0.080                    0.132       0.086

  Robust RMSEA                                               0.063
  90 Percent confidence interval - lower                     0.040
  90 Percent confidence interval - upper                     0.086
  P-value H_0: Robust RMSEA <= 0.050                         0.163
  P-value H_0: Robust RMSEA >= 0.080                         0.115

Standardized Root Mean Square Residual:

  SRMR                                           0.048       0.048

Parameter Estimates:

  Standard errors                             Sandwich
  Information bread                           Observed
  Observed information based on                Hessian

Latent Variables:
                   Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
  HP =~ esem
    HP1               0.302    0.052    5.802    0.000    0.302    0.466
    HP2               0.473    0.055    8.547    0.000    0.473    0.670
    HP3               0.603    0.059   10.138    0.000    0.603    0.647
    HP4               0.441    0.048    9.104    0.000    0.441    0.726
    HP5               0.485    0.056    8.592    0.000    0.485    0.634
```

```
    HP6               0.443    0.080    5.542    0.000    0.443    0.470
    OP1              -0.263    0.081   -3.262    0.001   -0.263   -0.230
    OP2              -0.081    0.085   -0.955    0.340   -0.081   -0.072
    OP3               0.319    0.074    4.320    0.000    0.319    0.269
    OP4              -0.118    0.085   -1.389    0.165   -0.118   -0.098
    OP5               0.091    0.062    1.480    0.139    0.091    0.086
    OP6              -0.036    0.057   -0.625    0.532   -0.036   -0.033
  OP =~ esem
    HP1              -0.040    0.033   -1.218    0.223   -0.040   -0.062
    HP2               0.047    0.041    1.154    0.248    0.047    0.067
    HP3               0.092    0.055    1.690    0.091    0.092    0.099
    HP4               0.004    0.032    0.121    0.904    0.004    0.006
    HP5              -0.071    0.043   -1.650    0.099   -0.071   -0.093
    HP6               0.027    0.081    0.339    0.735    0.027    0.029
    OP1               0.382    0.099    3.855    0.000    0.382    0.334
    OP2               0.409    0.100    4.078    0.000    0.409    0.363
    OP3               0.544    0.085    6.385    0.000    0.544    0.458
    OP4               0.595    0.110    5.383    0.000    0.595    0.493
    OP5               0.778    0.082    9.521    0.000    0.778    0.734
    OP6               0.901    0.072   12.526    0.000    0.901    0.833

Covariances:
                   Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
 .OP1 ~~
   .OP2               0.779    0.098    7.936    0.000    0.779    0.690
  HP ~~
    OP                0.265    0.080    3.321    0.001    0.265    0.265

Variances:
                   Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
   .HP1               0.334    0.042    7.980    0.000    0.334    0.794
   .HP2               0.260    0.045    5.755    0.000    0.260    0.522
   .HP3               0.468    0.089    5.238    0.000    0.468    0.538
   .HP4               0.173    0.033    5.257    0.000    0.173    0.470
   .HP5               0.364    0.068    5.326    0.000    0.364    0.621
   .HP6               0.684    0.110    6.196    0.000    0.684    0.771
   .OP1               1.144    0.105   10.920    0.000    1.144    0.876
   .OP2               1.113    0.098   11.401    0.000    1.113    0.877
   .OP3               0.923    0.095    9.741    0.000    0.923    0.653
   .OP4               1.127    0.149    7.578    0.000    1.127    0.773
   .OP5               0.472    0.093    5.074    0.000    0.472    0.420
   .OP6               0.373    0.084    4.430    0.000    0.373    0.319
    HP                1.000                              1.000    1.000
    OP                1.000                              1.000    1.000
```

Model Comparison

CFA and ESEM are **nested** (CFA is ESEM with all cross-loadings fixed to zero), so a scaled χ^2 difference test via `lavTestLRT()` is valid, in addition to fit index differences.

The null hypothesis is that the CFA model (more constrained) fits as well as the ESEM model (less constrained). A significant χ^2 difference test rejects this null, indicating that the added cross-loadings in ESEM significantly improve model fit.

A decrease in CFI > .01 between a less and more constrained model indicates meaningful misfit from the added constraints.

```
glance_custom.lavaan <- function(x, ...) {
  m <- lavaan::fitmeasures(x)
  tibble::tibble(
    cfi.robust   = m[["cfi.robust"]],
    tli.robust   = m[["tli.robust"]],
    rmsea.robust = m[["rmsea.robust"]]
  )
}

gm <- data.frame(
  raw   = c("nobs", "statistic", "chi2.df", "p.chi2",
            "cfi.robust", "tli.robust", "rmsea.robust", "srmr"),
  clean = c("N", "Chisq", "df", "p(Chisq)",
            "CFI*", "TLI*", "RMSEA*", "SRMR"),
  fmt   = c(0, 2, 0, 3, 3, 3, 3, 3)
)
modelsummary::msummary(
    list("CFA" = fit_cfa, "ESEM" = fit_esem),
    fmt = fmt_decimal(digits = 2),
    estimate = "{estimate} [{conf.low}, {conf.high}]",
    statistic = NULL,
    metrics = "all",
    gof_map = gm
)
```

	CFA	ESEM
HP = HP1	0.28 [0.18, 0.39]	0.30 [0.20, 0.40]
HP = HP2	0.49 [0.39, 0.60]	0.47 [0.36, 0.58]
HP = HP3	0.65 [0.54, 0.75]	0.60 [0.49, 0.72]
HP = HP4	0.44 [0.35, 0.53]	0.44 [0.35, 0.54]
HP = HP5	0.45 [0.33, 0.56]	0.49 [0.37, 0.60]
HP = HP6	0.46 [0.32, 0.59]	0.44 [0.29, 0.60]
OP = OP1	0.27 [0.08, 0.46]	0.38 [0.19, 0.58]

	CFA	ESEM
OP = OP2	0.37 [0.19, 0.55]	0.41 [0.21, 0.61]
OP = OP3	0.66 [0.50, 0.81]	0.54 [0.38, 0.71]
OP = OP4	0.55 [0.35, 0.75]	0.59 [0.38, 0.81]
OP = OP5	0.82 [0.67, 0.97]	0.78 [0.62, 0.94]
OP = OP6	0.87 [0.74, 1.01]	0.90 [0.76, 1.04]
OP1 OP2	0.82 [0.63, 1.01]	0.78 [0.59, 0.97]
HP OP	0.34 [0.19, 0.50]	0.26 [0.11, 0.42]
HP = OP1		-0.26 [-0.42, -0.10]
HP = OP2		-0.08 [-0.25, 0.09]
HP = OP3		0.32 [0.17, 0.46]
HP = OP4		-0.12 [-0.29, 0.05]
HP = OP5		0.09 [-0.03, 0.21]
HP = OP6		-0.04 [-0.15, 0.08]
OP = HP1		-0.04 [-0.11, 0.02]
OP = HP2		0.05 [-0.03, 0.13]
OP = HP3		0.09 [-0.01, 0.20]
OP = HP4		0.00 [-0.06, 0.07]
OP = HP5		-0.07 [-0.16, 0.01]
OP = HP6		0.03 [-0.13, 0.19]
N	210	210
Chisq	114.94	79.09
df	52	42
p(Chisq)	0.000	0.000
CFI*	0.913	0.950
TLI*	0.890	0.921
RMSEA*	0.075	0.063
SRMR	0.080	0.048

```
# Factor correlations from CFA and ESEM
cfa_cor  <- lavInspect(fit_cfa,  "cor.lv")
esem_cor <- lavInspect(fit_esem, "cor.lv")
data.frame(
    Model = c("CFA", "ESEM"),
    r_HP_OP = c(round(cfa_cor["HP", "OP"], 3),
```

```
                 round(esem_cor["HP", "OP"], 3))
)
```

```
  Model r_HP_OP
1   CFA   0.345
2  ESEM   0.265
```

> **Note**
>
> **Interpretation**: If ESEM provides substantially better fit (ΔCFI > .01, ΔRMSEA meaningful) and the freed cross-loadings are theoretically sensible, prefer ESEM. If fit is similar and cross-loadings are negligible, retain the more parsimonious CFA.

References

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